

## Topics

- Algebra of sets
- Complements of open and closed
- Union + intersection of open + closed
- Closure
- Density
  - Sup and inf are in the closure

Def  $E \subset X$  is closed if every limit point of  $E$  is in  $E$ . That is,  $E' \subset E$ .

Examples:

1) Any finite subset of a metric space is closed.

2)  $[a, b] \subset \mathbb{R}$  is closed.

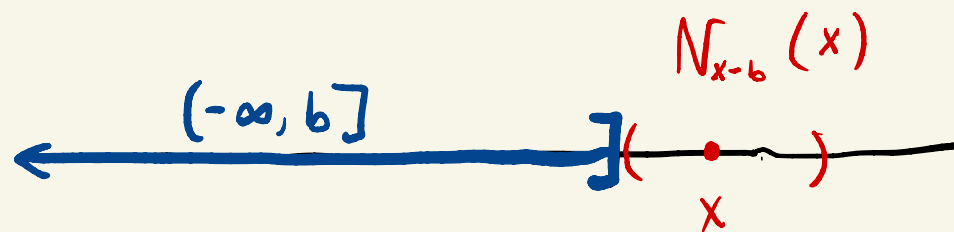
3)  $(-\infty, b] = \{x \in \mathbb{R} \mid x \leq b\}$  is closed. So is  $[a, \infty)$ .

proof of 3: We show every  $x \notin (-\infty, b]$  is not a limit point.

Suppose  $x > b$ . Let  $y \in N_{x-b}(x)$ .

$$y > x - (x - b) = b$$

So  $y \notin (-\infty, b]$ , so  $N_{x-b}(x) \cap [a, b] = \emptyset$ , so  $x$  is not a limit point.



Examples Think of subsets of  $\mathbb{R}$  which are...

open + closed (clopen)

$\mathbb{R}$

$\emptyset$

open + not closed

$(a, b)$

$(a, b) \cup (b', a')$

$\bigcup_{n \in \mathbb{N}} (n, n + \frac{1}{2})$

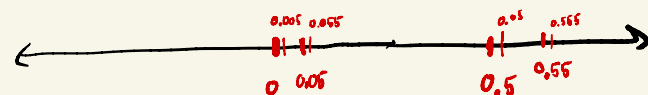
closed + not open

•  $[a, b]$

• Finite sets  $\{r_1, \dots, r_n\}$

•  $\mathbb{Z}$

•  $\{r \in \mathbb{R} \mid \text{decimal expansion contains only 0s and 5s}\}$



not open + not closed

$[a, b)$

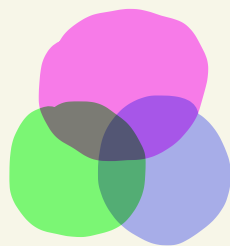
$\mathbb{Q}$

$\mathbb{R} \setminus \mathbb{Q}$

# Algebra of subsets

$\cap$  and  $\cup$  are like  $\cdot$  and  $+$ .

Let  $A, B, C \subseteq S$



$$A \cap B = B \cap A, \quad A \cup B = B \cup A$$

Distributivity

$$(A \cap B) \cap C = A \cap (B \cap C), \quad (A \cup B) \cup C = A \cup (B \cup C) \quad \text{Associativity}$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Distributivity

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Extra distributivity!

$\emptyset$  is the unit for  $\cup$

$S$  is the unit for  $\cap$

$\mathcal{P}(S)$  has another structure: complements.

Def Let  $A \subseteq S$ .  $A^c = \{x \in S \mid x \notin A\}$

$A \mapsto A^c$  exchanges  $\cup$  and  $\cap$

$$\cdot (A \cup B)^c = A^c \cap B^c$$

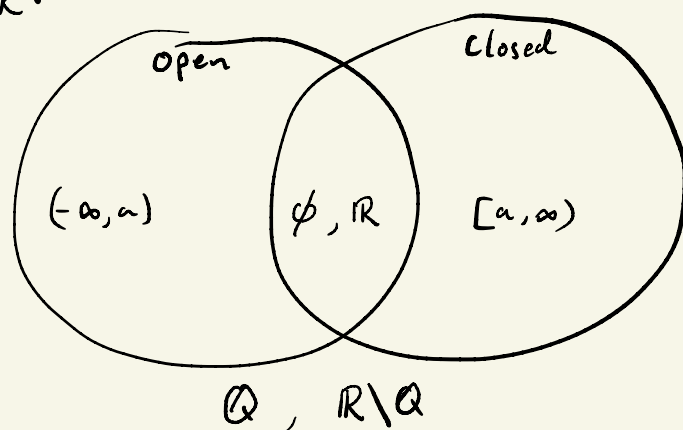
$$\cdot (A \cap B)^c = A^c \cup B^c$$

Theorem (2.23) Let  $X$  be a metric space, and  $E \subset X$  be a subset.

a)  $E$  open  $\Rightarrow E^c$  closed.

b)  $E$  closed  $\Rightarrow E^c$  open.

Examples in  $\mathbb{R}$ :



proof: Suppose  $E$  is open. Let  $x$  be limit point of  $E^c$ .

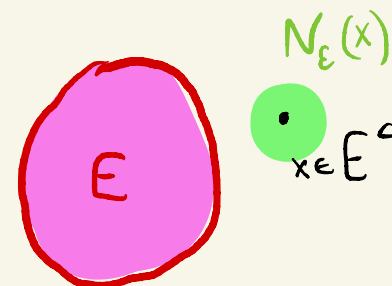
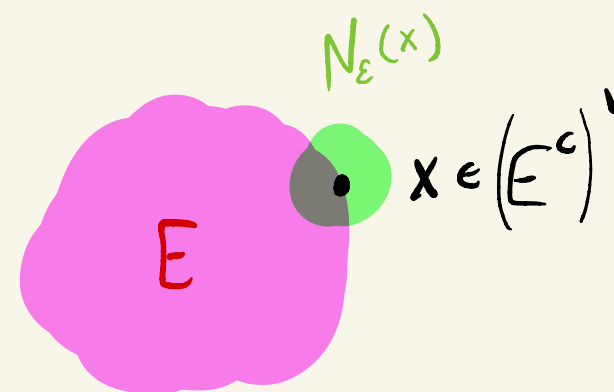
Every nbhd  $N$  of  $x$  intersects  $E^c$ , so  $x$  is not in  $E$ . Therefore  $x$  is not

interior point of  $E$ , so  $x \in E^c$ .

Suppose  $E$  closed. Let  $x \in E^c$ .

$E$  closed, so  $x$  is not limit pt of  $E$ . This means

$\exists$  nbhd of  $x$  that doesn't intersect  $E$ , thus  $x$  is in  $E^c$ .



## Theorem (2.24)

- (a) A union of open sets is open.
- (b) An intersection of closed sets is closed
- (c) A finite intersection of open sets is open
- (d) A finite union of closed sets is closed

proof

(a) Let  $E_\alpha$  be open.

Let  $x \in \bigcup_\alpha E_\alpha$

$\Rightarrow x \in E_\alpha$  for some  $\alpha$

$\Rightarrow x$  interior pt of  $E_\alpha$

$\Rightarrow x$  interior pt. of  $\bigcup_\alpha E_\alpha$ .

(b) Let  $F_\alpha$  be closed

$\Rightarrow F_\alpha^c$  are open

$\Rightarrow \bigcup_\alpha F_\alpha^c$  is open

$\Rightarrow \left(\bigcup_\alpha F_\alpha^c\right)^c = \bigcap_\alpha F_\alpha$  is open

(c) Let  $E_1, \dots, E_n$  be open.

Let  $x \in E_1 \cap \dots \cap E_n$

$\Rightarrow N_{r_i}(x) \subset E_i$  for some  $r_i > 0$  for each  $i$

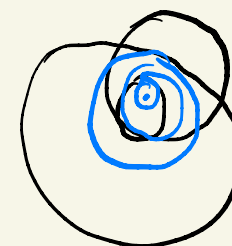
$\Rightarrow N_{\min\{r_i\}} \subset E_1 \cap \dots \cap E_n$

(d) Let  $E_1, \dots, E_n$  be closed

$\Rightarrow E_1^c, \dots, E_n^c$  open

$\Rightarrow E_1^c \cap \dots \cap E_n^c$  open

$\Rightarrow \left(E_1^c \cap \dots \cap E_n^c\right)^c = E_1 \cup \dots \cup E_n$  closed



Example:  $\bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}$  Finiteness is essential.